

Production Monitoring and Auto-Healing AWS Infrastructure

Overview

This project demonstrates a production-grade AWS architecture with monitoring, auto scaling, and automated incident recovery using CloudWatch, SNS, ALB, and EC2.

Architecture

Users -> Application Load Balancer -> Auto Scaling Group -> EC2 Instances -> CloudWatch -> SNS Alerts

AWS Services Used

EC2, Application Load Balancer, Auto Scaling Group, CloudWatch, SNS, IAM, Security Groups, Target Groups

Implementation

React dashboard deployed on EC2 behind Nginx. AMI created for scaling. Load Balancer distributes traffic across instances.

Monitoring and Alerts

CloudWatch tracks CPU utilization, network metrics, and ALB health. SNS sends email alerts for threshold breaches.

Auto Scaling

Auto Scaling Group maintains 2 to 4 instances. Scaling triggered based on CPU utilization target of 60 percent.

Incident Simulation

CPU stress test and instance termination were used to simulate production failures. System automatically recovered using Auto Scaling.

Outcome

High availability infrastructure with automated recovery, monitoring, and alerting suitable for production workloads.